# Mikhail Kornev

PhD in Mathematics

Steklov Mathematical Institute of RAS
HSE University

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## RESEARCH AREAS

Geometry and Topology

## APPOINTMENTS

| | |
|---|---|
| Sep. 2022 – | Research assistant at Steklov Mathematical Institute of RAS<br>Department of Geometry and Topology |
| Mar. 2026 – | Research assistant at HSE University<br>International Laboratory for Mirror Symmetry and Automorphic Forms |
| Sep. 2021 – 2025 | Research assistant at HSE University<br>Laboratory of Algebraic Topology and Applications |

## EDUCATION

| | |
|---|---|
| 2026 | PhD in Mathematics, Steklov Mathematical Institute of RAS<br>Doctoral studies: 2022 – 2026<br>Department of Geometry and Topology |
| Sep. 2016 – May 2022 | Specialist degree Diploma with honors in Mathematics (equivalent to M.S.)<br>Lomonosov Moscow State University |

## RESEARCH PAPERS

### Mathematics

| | |
|---|---|
| Jul. 2026 | Buchstaber V.M., Kornev M.I.,<br>*Algebraic n-valued groups, cubics, and Wendt matrices*<br>Matematicheskii Sbornik, Volume 217, Number 8 (2026)<br>https://doi.org/10.4213/sm10344 |
| Apr. 2026 | Kornev M.I.,<br>*Buchstaber, Ochanine, Krichever, and Witten genera*<br>Journal of Geometry and Physics, Volume 226, Number 105855 (2026)<br>https://doi.org/10.1016/j.geomphys.2026.105855 |
| Jan. 2026 | Buchstaber V.M., Kornev M.I.<br>*n-Valued Monoids on $\mathbb{C}P^1$ and Discriminants*<br>Russian Mathematical Surveys, Volume 81, Issue 1(487) (2026), 191–193<br>https://doi.org/10.4213/rm10280e |

Jan. 2026 — Kornev M.I.
*On the Classification of n-Valued Monoids and Groups of Order* 3
Mat. Zametki, Volume 119, Issue 2 (2026), 244–252
https://doi.org/10.4213/mzm14803

Jan. 2024 — Kornev M.I.
*n-Valued Coset Groups and Dynamics*
Mathematical Notes, Volume 116, Issue 1 (2024), 66–76
https://doi.org/10.4213/mzm14243

Sep. 2023 — Kornev M.I.
*Homology Spheres*
Transactions of the Moscow Mathematical Society, Volume 84 (2023), 197–239
https://doi.org/10.1090/mosc/352

### Mathematics and Music

Sep. 2025 — *The Technique of Musical Icosahedra*
Music Academy, no. 3, 2025, pp. 64–83
https://doi.org/10.34690/479

## PREPRINTS

Sep. 2026 — Buchstaber V.M., Kornev M.I.
*Logarithm of the Universal Two-Valued Formal Group*
arXiv:2609.15878

Apr. 2026 — Buchstaber V.M., Kornev M.I., Rubtsov V.N.
*Two-Valued Groups, Chazy Equation, Dubrovin-Frobenius Structures, and QYBE*
arXiv:2604.26641

Mar. 2026 — Kornev M.I.,
*Buchstaber, Ochanine, Krichever, and Witten genera*
arXiv:2603.21118

Oct. 2025 — Buchstaber V.M., Kornev M.I.
*Algebraic n-Valued Monoids on $\mathbb{C}P^1$, Discriminants and Projective Duality*
arXiv:2510.14010

Aug. 2025 — Kornev M.I.
*On the Classification of n-Valued Monoids and Groups of Order* 3
arXiv:2508.04454

May 2025 — Buchstaber V.M., Kornev M.I.
*n-Valued Groups, Kronecker Sums, and Wendt's Matrices*
arXiv:2505.04296

Jan 2024 — Kornev M.I.
*n*-valued Coset Groups and Dynamics
arXiv:2401.12718

## CONFERENCE TALKS

Oct. 8, 2026 — *Two-Valued Groups, Chazy Equation, Dubrovin-Frobenius Structures, and QYBE*
International Conference "Integrable Systems and Quantum Theory", dedicated to the 80th anniversary of A. K. Pogrebkov,
Saint Petersburg State University, Saint Petersburg, Russia

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| May. 2026 | *Buchstaber Genus and Formal Groups*<br>Conference "Algebraic Topology, Group Actions, and Combinatorics," Sirius, Sochi, Russia |
| Aug. 2025 | *n-Valued Groups, Kronecker Sums, and Wendt Matrices*<br>joint work with Victor Buchstaber<br>5th Conference of Mathematical Centers of Russia, Krasnoyarsk, Russia |
| May 2024 | *Homology Spheres, Acyclic Groups and Kan-Thurston Theorem*<br>Conference "From Analysis to Homotopy Theory" , University of Greifswald, Greifswald, Germany |
| Apr. 2023 | *Dynamics and multivalued groups*<br>Conference "SIMC Youth Race", Steklov Mathematical Institute of RAS, Moscow, Russia |
| Oct. 2022 | *Inconcretness of homotopy categories*<br>International School "Toric topology, combinatorics and data analysis", Euler International Mathematical Institute, Saint Petersburg, Russia |
| Apr. 2022 | *Acyclic groups and spaces*<br>29th International Scientific Conference for Undergraduate and Graduate Students and Young Scientists "Lomonosov", Moscow, Russia |
| Nov. 2021 | *Acyclic groups and Kan-Thurston theorem*, International School "Toric Topology and Combinatoric", Sirius Mathematics Center, Sochi, Russia |

## SEMINAR TALKS

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| Sep. 22, 29, 2026 | *Buchstaber, Witten, Krichever, and Ochanine Genera* (2 talks)<br>Seminar "Automorphic Forms and Their Applications",<br>Faculty of Mathematics, HSE University, Moscow, Russia |
| May 2026 | *Buchstaber genus and formal groups*<br>Seminar "Toric Topology, Combinatorics, and Homotopy Theory"<br>jointly with the Laboratory of Topology and Complex Networks of the Faculty of Mathematics, HSE University,<br>Moscow Center for Continuous Mathematical Education (MCCME), Moscow, Russia |
| Dec. 2025 | *Fermat hypersurfaces, projective duality, and algebraic n-valued monoids*<br>Seminar of the Department of Geometry and Topology "Geometry, Topology and Mathematical Physics" (Novikov Seminar),<br>Steklov Mathematical Institute of RAS, Moscow, Russia |
| Dec. 2025 | *Discrete and Algebraic n-Valued Groups*<br>Seminar on Discrete Geometry and Geometry of Numbers,<br>Moscow State University, Moscow, Russia |
| Nov. 2025 | *n-Valued Algebraic Monoids, Cubics, and Discriminants*<br>The Seminar of International Laboratory of Algebraic Topology and Its Applications, HSE University, Moscow, Russia |
| May 2025 | *n-Valued Groups, Kronecker Sums, and Wendt Matrices*<br>Joint work with Victor Buchstaber<br>Seminar of the Department of Geometry and Topology "Geometry, Topology and Mathematical Physics",<br>Steklov Mathematical Institute of RAS (Novikov Seminar), Moscow, Russia |
| Apr. 2023 | *Acyclic groups and Kan-Thurston theorem* |

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| | Seminar by I. G. Lysenok and A. L. Talambutsa "Seminar on geometric group theory", Steklov Mathematical Institute of RAS, Moscow, Russia |
| Dec. 2022 | *Fundamental groups of homology spheres*<br>Seminar "Geometry, Topology and their applications" at Sobolev Institute of Mathematics,<br>Online |

## GRANTS AND AWARDS

### Prizes, fellowships and grants

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| 2021 – 2023 | *The Russian Science Foundation Grant No. 23-11-00143* |
| 2019 | *Winner of the olympiad on Partial Differential Equations*<br>Moscow State University |
| 2018 | *Winner of the olympiad on Ordinary Differential Equations*<br>Moscow State University |

## STUDY SEMINAR TALKS

### Texas Tech Quantum Homotopy Seminar

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| Dec. 2024 | *An Introduction to Cartan Geometries* |
| Mar. 2024 | *Euler-Lagrange dynamics via the $\infty$-jet bundle* (2 talks) |

### Seminar On Homotopy Theory and Applications at Independent University of Moscow

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| Dec. 2023 | *Principal $\infty$-bundles* |
| Apr. – May 2023 | *A mini-course on the $\infty$-category of motivic coarse spaces* (4 talks) |
| Apr. 2023 | *Elements of unstable motivic homotopy theory* |
| Feb. – Apr. 2023 | *A mini-course on $\infty$-cosmoi* (4 talks) |
| Dec. 2022 | *Applications of Reedy theory* |
| Nov. 2022 | *Weighted (co-)limits, Gambino's theorem, Reedy model structures* |
| Nov. 2022 | *Homotopically final functors and examples of calculations with homotopy (co-)limits* |
| Nov. 2022 | *Nerve—realization adjunction and bar construction* |
| Apr. 2022 | *A series of 4 talks on sheaves of sets* |
| Apr. 2022 | *Kan extensions. Additional topics* |
| Oct. 2021 | *The categorical meaning of the Kan-Thurston theorem* |
| Jul. – Aug. 2021 | *Quillen's theorem, a closed model structure on $sSet$* |

### Study And Scientific Seminar "Geometry And Topology" at Moscow State University

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| Sep. 2020 | *Andreev's theorem on hyperbolic polyhedra* |
| Nov. 2019 | *The intersection form of 4-manifolds* |
| May 2019 | *Nilmanifolds and Massey product in bigraded cohomology rings* |

## CONFERENCES, WORKSHOPS AND SCHOOLS

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| May 2024 | *Conference "From Analysis to Homotopy Theory"*<br>University of Greifswald, Greifswald, Germany |
| Jul. 2023 | *Summer School "Algebra and Geometry"*<br>HSE University, Steklov Mathematical Institute of RAS, Suzdal, Russia |
| Jul. 2023 | *Summer school on non-commutative and derived geometry of matrix factorisations*<br>Centre of Pure Mathematics, Moscow Institute of Physics and Technology, Dolgoprudny, Russia |
| Jun. 2023 | *Conference on algebra, algebraic geometry, and number theory on the occasion of academician Igor Rostislavovich Shafarevich 100th birthday*<br>Steklov Mathematical Institute of RAS, Moscow, Russia |
| Apr. 2023 | *Conference "Geometry, topology and mathematical physics" on the occasion of the 85th anniversary of S. P. Novikov and the 80th anniversary of V. M. Buchstaber*<br>Steklov Mathematical Institute of RAS, Moscow, Russia |
| Mar. 2023 | *Conference "SIMC Youth Race"*<br>Steklov Mathematical Institute of RAS, Moscow |
| Oct. 2022 | *International School "Toric topology, combinatorics and data analysis"*<br>Euler International Mathematical Institute, Saint Petersburg, Russia |
| Jul. 2022 | *Summer School "Algebra and Geometry"*<br>HSE University, Steklov Mathematical Institute of RAS, Suzdal, Russia |
| Apr. 2022 | *29th International Scientific Conference for Undergraduate and Graduate Students and Young Scientists "Lomonosov"*<br>Lomonosov Moscow State University, Moscow, Russia |
| Nov. 2021 | *International School "Toric Topology and Combinatorics"*<br>HSE University, Sirius Mathematics Center, Sochi, Russia |
| Nov. 2020 | *Topology and Geometry of Group Actions*<br>HSE University, online |
| Nov. 2019 | 7th Summer School On Geometrical Methods Of Mathematical Physiques<br>Landau Institute for Theoretical Physics, Lomonosov Moscow State University, Chernogolovka, Russia |
| Nov. 2019 | *Geometry, Topology And Mathematical Physics*<br>Landau Institute for Theoretical Physics, Lomonosov Moscow State University, Chernogolovka, Russia |
| Nov. 2018 | *Geometry, Topology And Mathematical Physics*<br>Landau Institute for Theoretical Physics, Lomonosov Moscow State University, Chernogolovka, Russia |
| Jul. 2018 | *Summer school "Modern Mathematics"*<br>HSE University, Lomonosov Moscow State University, Independent University of Moscow, Dubna, Russia |
| Jul. 2017 | *Summer school "Modern Mathematics"*<br>HSE University, Lomonosov Moscow State University, Independent University of Moscow, Dubna, Russia |

## TEACHING

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| May 2022 – 2024 | *Teaching various courses (e. g., Algebraic Geometry, Algebraic Topology, Abstract Algebra, Differential Geometry, Linear Algebra, Probability Theory, Discrete Mathematics, Calculus, etc.) on Profi*<br>`https://profi.ru/profile/KornevMI3/` |
| Sep. 2016 – 2022 | *Little Mech-Math courses*<br>An olympiad Math class at Lomonosov Moscow State University<br>`http://mmmf.msu.ru/about/person.html` |

## SERVICE

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| June 2021 – 2024 | *Co-organization of the study seminar "Homotopy Theory and Applications" at Independent University of Moscow*<br>`https://researchseminars.org/seminar/HomotopyTheoryAndApps` |

## REFERENCES

*Victor Buchstaber*,
Steklov Mathematical Institute of RAS
`buchstab@mi-ras.ru`

## LANGUAGES

*Russian* (native)

*English* (fluent)

*French* (intermediate)

*Latin* (basic)